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Mainlobe Deceptive Interference Suppression With Planar Frequency Diverse Array MIMO Radar

Mengdi Zhang, Jiahao Lu, Jingwei Xu, and Shiyin Li

***Abstract*—Frequency diversity array (FDA) is capable of suppressing mainlobe deceptive interference with delays exceeding one pulse relative to the true target, known as inter-pulse interference. However, current FDA-based interference suppression methods are ineffective when both inter-pulse and intra-pulse interferences coexist. To this end, we propose a two-stage mainlobe deceptive interference suppression method with planar FDA multiple-input-multiple-output (PFDA-MIMO) radar: 1) Horizontal transmit spatial frequency compensation (HTSFC) is applied using the range information of each range bin to differentiate between inter-pulse interferences and target. Subsequently, data-independent beamforming is performed to suppress inter-pulse interferences in horizontal spatial frequency domain; 2) To distinguish the target from intra-pulse interferences, vertical transmit spatial frequency compensation (VTSFC) is applied utilizing the initial range of the range bin containing target. Afterwards, an adaptive beamforming (ABF) approach based on transmit-receive spatial smoothing (TRSS) is applied to suppress intra-pulse interference while avoiding target cancellation. Furthermore, the frequency increments in both rows and columns of PFDA are elaborately designed to effectively distinguish between true target and interferences. Extensive simulations are conducted to evaluate the excellent suppression effect of the proposed scheme against mainlobe deceptive interference.**

***Index Terms*—Planar frequency diversity array multiple-input multiple-output (PFDA-MIMO), two-dimensional (2-D) range degrees of freedom (DOFs), mainlobe deceptive interference suppression, intra-pulse interference, inter-pulse interference, spatial frequency domain.**

I. Introduction

THE modern battlefield electronic environment is becoming increasingly complex, posing serious challenges to the normal function of radar systems because of advancements in jamming technology. Among various types of jammers, the deceptive jammer has the capability to intercept radar signals and employ digital techniques to retrieve, replicate, and parametrically modulate radar waveforms stored in digital radio frequency memory (DRFM) [1], [2], [3]. This process, known as "intercept-modulate-retransmit", generates numerous false targets that closely resemble the true target. Conventional phased array (PA) radars can effectively suppress deceptive interference from sidelobes using adaptive interference suppression technique. However, when the interferences are distributed in the mainlobe, the target detection performance is compromised, significantly limiting the ability of the radar system to acquire target information [4].

Recently, frequency diverse array (FDA) radar has garnered significant attention from researchers [5]-[9]. To leverage range degrees of freedom (DOFs) from the transmitting end, integrating FDA with multiple-input-multiple-output (MIMO) emerges as an indispensable choice [10]-[16], offering significant potential in suppressing mainlobe deceptive interference [17]-[24]. In [17], the signal model of deceptive interference with FDA-MIMO radar is firstly derived. Utilizing the additional range DOFs, the mainlobe deceptive jamming suppression is achieved through transmit-receive adaptive beamforming (ABF). In [18], an improved three-dimensional (3-D) joint domain localized space-time adaptive processing (STAP) method is introduced, addressing the severe performance degradation caused by dense false targets. To address the decreased interference suppression performance due to interference violating the principle of independent and identical distribution, [19] introduces a robust ABF approach using direct data domain (DDD) technique. Considering the modulation delay of false targets, [20] proposes a sample selection approach using subspace projection to cope with the issue of pseudo-random distribution of false targets. To ensure the suppression of false targets within a specified range, a preset broadened null beamformer is designed in [21]. By employing the minimum redundancy array (MRA) in the transmitting end of FDA-MIMO, [22] introduces an interference suppression method capable of suppressing the number of interferences exceeding system DOFs. However, the aforementioned methods are only effective in suppressing the false targets delayed by more than one pulse relative to the true target (known as inter-pulse interference). To suppress both the inter-pulse interference and the interference that shares the same number of delayed pulses but different range bin with the true target, [23] proposes a new receive delay array (RDA). However, the time delay across the receive elements is on the nanosecond scale, posing a significant challenge for implementation. Additionally, this method is only effective when employing linear frequency modulated (LFM) signals. Advanced DRFM technology can generate intra-pulse interference with a brief time delay, exhibiting identical delayed pulses number and range bin with the true target. In

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[24], a novel FDA-MIMO framework with a dual-frequency increment (DFI) scheme is proposed to combat intra-pulse interference. However, this method concentrates on scenarios where only intra-pulse interference exists. Notice that these methods are invalid when intra-pulse and inter-pulse interferences coexist.

The planar frequency diverse array (PFDA) introduces frequency offsets in both rows and columns, thereby providing range information in both vertical and horizontal dimensions. However, there is limited research currently available on PFDA [25]-[30]. [25] firstly extends the beamforming theory of FDA to planar array with the frequency offsets along both rows and columns. In [26], a 3-D ABF method is applied to PFDA. However, the relationship between frequency offsets and beamforming performance have not been fully explored. In [27], the transmitting beampattern characteristics of the coherent PFDA are analyzed, focusing on its 3-D space scanning characteristics and the temporal variation of the mainlobe. A method based on static non-linear frequency offset and time modulation weight is proposed in [28], which concentrates energy on the defined 3-D acquisition area. To address serious Doppler spread in FDA, a technique with PFDA is proposed for moving target detection, achieving a reduction of at least 3dB in the input signal-to-noise ratio (SNR) [29]. In [30], a multi-target 3-D localization approach with PFDA-MIMO using tensor subspace is introduced. However, the unique two-dimensional (2-D) range information of PFDA has not been fully explored in the aforementioned papers.

In this study, a novel PFDA-MIMO framework is constructed to cope with a complex mainlobe deceptive interference scenario, involving three types of interference. The first type is inter-pulse interference, which has a delay of at least one pulse later than the true target. The second type shares the same number of delayed pulses but lags behind the range bin of the target. The third type is intra-pulse interference, also referred to fast repeater deception interference, which shares both the same range bin and the same number of delayed pulses as the true target. Firstly, we establish the signal model of PFDA-MIMO radar and design multiple matched filters to recover the 2-D range DOFs. Then, a two-stage beamforming method is proposed to suppress mainlobe deceptive interference. 1) In horizontal transmit spatial frequency domain, echoes are compensated in a range-bin-by-range-bin mode, enabling the clear differentiation of inter-pulse interferences. Subsequently, data-independent beamforming in horizontal transmit-receive spatial frequency domain is performed to suppress inter-pulse interferences. 2) In vertical transmit spatial frequency domain, echoes after first-stage beamforming are compensated using the initial range of the range bin associated with target, facilitating the distinction between intra-pulse interferences and the true target. To suppress intra-pulse interferences, vertical transmit-receive ABF based on transmit-receive spatial smooth (TRSS) is implemented. Additionally, meticulous design of both row and column frequency offsets is executed to effectively separate the false targets. The primary achievements of this work are briefly stated as follows:

- The universal signal model of PFDA-MIMO radar is established, where the controllable range DOFs are 2-D compared to that of FDA-MIMO radar.
- A two-stage beamforming method with PFDA-MIMO radar is proposed, capable of simultaneously suppressing both mainlobe inter-pulse and intra-pulse interferences. The proposed method demonstrates significantly superior suppression performance compared to existing methods, with a performance loss not exceeding 3 dB relative to the ideal case.

The remainder of this paper is structured like following. Section II introduces the target and deceptive interference signal models of PFDA-MIMO radar, followed by the proposed mainlobe deceptive interference suppression method with PFDA-MIMO radar in Section III. Simulation results that corroborate the superior performance of PFDA-MIMO radar for suppressing deceptive interference are illustrated in Section IV. Finally, Section V provides the concluding remarks of this study.

Notations: $(\cdot)^*$, $(\cdot)^{\mathrm{T}}$, $(\cdot)^{-1}$ and $(\cdot)^{\mathrm{H}}$ represent the conjugate, transpose, inverse and conjugate transpose, respectively. $\otimes$ and $\odot$ are the Kronecker and Hadamard product operators, respectively. $\mathbb{C}$ and $\mathbb{N}$ denote the sets of complex numbers and natural numbers, respectively. $\mathbf{0}_M$ and $\mathbf{1}_M$ are the $M\times1$ vectors with all elements being 0 and 1, respectively. $\mathbf{I}_M$ denotes the $M\times M$ identity matrix.

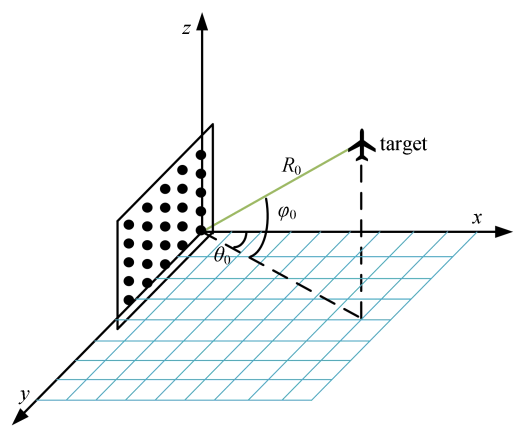

Fig. 1. Geometry of PFDA-MIMO radar.

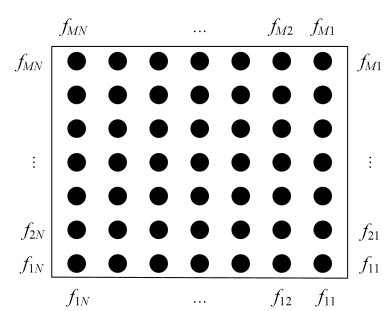

Fig. 2. Structure of the PFDA.

II. Signal Model of PFDA-MIMO Radar

Fig. 1 shows an airborne PFDA-MIMO radar in 3-D cartesian coordinate system. Both the transmitting and receiving array adopt identical planar array. Fig. 2 illustrates the structure of PFDA, where the number of rows and columns

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are M and N, respectively. The interelement spacing of the planar array is d in both rows and columns. The transmitting element located in the mth (m=1,2,...,M) row and nth (n=1,2,...,N) column is referred as (m,n) transmitting element. The traditional FDA system employs a slightly frequency increment along either rows or columns. Unlike FDA, PFDA exhibits simultaneous variation of the carrier frequency along both rows and columns. Thus, the carrier frequency of the (m,n) transmitting element can be expressed as

$$f_{mn} = f_{11} + (m-1)\Delta f_{row} + (n-1)\Delta f_{col} \tag{1}$$

where $f_{11}=f_0$ is the reference carrier frequency. Δf_{row} and Δf_{col} represent the frequency increments between adjacent rows and columns, respectively. Note that Δf_{row} and Δf_{col} are much smaller than f_0.

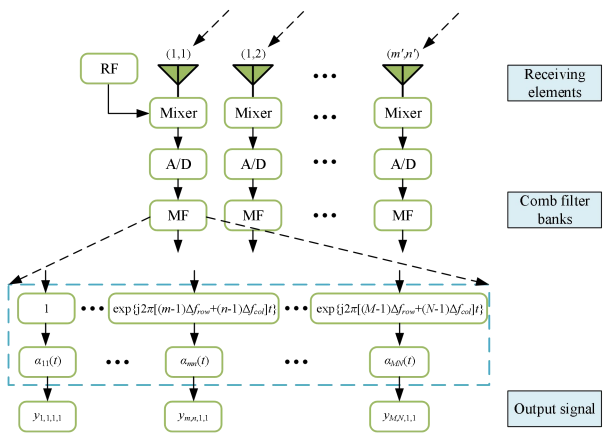

Fig. 3. Multiple matched filters system of PFDA-MIMO radar.

A. Signal model of target

The transmitting signal of the (m,n) transmitting element is represented as

$$s_{mn}(t) = \sqrt{\frac{E}{MN}}\alpha_{mn}(t)e^{j2\pi f_{mn}t}, \qquad 0 \le t \le T_w \tag{2}$$

where E, T_w and $\alpha_{mn}(t)$ are the total transmitting energy, the pulse width of signal and the complex envelope corresponding to the (m,n) transmitting element, respectively. It is assumed that the complex envelopes satisfy ideal orthogonal conditions with one another:

$$\int_{T_w} \alpha_{m_1 n_1}(t)\alpha^*_{m_2 n_2}(t-\tau)dt = 0, \quad m_1 \ne m_2 \cup n_1 \ne n_2, \forall \tau \tag{3}$$

where τ is an arbitrary time delay. The design of orthogonal waveforms remains a conspicuous and challenging research area [31-35]. In weak clutter and interference scenarios, the target can be treated as a point source, making the assumption of ideal orthogonal waveforms reasonable and consistent with previous works [4, 13, 14, 36, 37].

In this planar array, the element in the right lower corner is selected as the reference element. Assuming that there is a far-field point target P_0 located at (R_0,θ_0,φ_0), where R_0, θ_0, φ_0 represent the slant range, azimuth angle and elevation angle from P_0 to the reference element, respectively. The receiving element corresponding to the m'th row and n'th column is termed as (m',n') receiving element. The echo received by the (m',n') receiving element is represented as

$$x_{m'n'}(t) = \rho\sum_{m=1}^{M}\sum_{n=1}^{N}\alpha_{mn}(t-\tau_0)e^{j2\pi f_{mn}(t-\tau_{mn}-\tau_{m'n'})} \tag{4}$$

where ρ denotes the target complex reflection coefficient. $\tau_0=2R_0/c$ is the common two-way time delay, where c represents the speed of light. τ_{mn} and $\tau_{m'n'}$ denote the propagation delay of (m,n) transmitting element and the receiving delay of (m',n') receiving element, respectively, which can be specifically represented as

$$\tau_{mn} = \frac{R_0 - d(m-1)\sin(\varphi_0) - d(n-1)\sin(\theta_0)\cos(\varphi_0)}{c} \tag{5}$$

$$\tau_{m'n'} = \frac{R_0 - d(m'-1)\sin(\varphi_0) - d(n'-1)\sin(\theta_0)\cos(\varphi_0)}{c} \tag{6}$$

To recover the DOFs in transmitter, down-conversion and matched filtering are carried out on the echoes of each receiving channel [38], as shown in Fig. 3. Note that each receiving channel contains $M\times N$ matched filters. Then, the echo corresponding to the (m,n) transmitting element and the (m',n') receiving element is expressed as

$$y_{m,n,m',n'} = \rho e^{\frac{-j4\pi f_{mn}R_0}{c}} e^{\frac{j2\pi f_{mn}(d(m-1)\sin(\varphi_0)+d(n-1)\sin(\theta_0)\cos(\varphi_0))}{c}} \times e^{\frac{j2\pi f_{mn}(d(m'-1)\sin(\varphi_0)+d(n'-1)\sin(\theta_0)\cos(\varphi_0))}{c}} \tag{7}$$

Due to $\Delta f_{row} \ll f_0$ and $\Delta f_{col} \ll f_0$, Eq. (7) can be further expanded and approximated as

$$y_{m,n,m',n'} = \xi e^{j2\pi\left(-2R_0\frac{(m-1)\Delta f_{row}}{c} + \frac{(m-1)d\sin(\varphi_0)}{\lambda_0}\right)} e^{j2\pi\left(-2R_0\frac{(n-1)\Delta f_{col}}{c} + \frac{(n-1)d\sin(\theta_0)\cos(\varphi_0)}{\lambda_0}\right)} \times e^{j2\pi\frac{(m'-1)d\sin(\varphi_0)}{\lambda_0}} e^{j2\pi\frac{(n'-1)d\sin(\theta_0)\cos(\varphi_0)}{\lambda_0}} \tag{8}$$

where $\xi=\rho\exp\{-j2\pi f_0\tau_0\}$. $\lambda_0=c/f_0$ is the radar wavelength. Therefore, the received snapshots of PFDA-MIMO radar is represented as

$$\begin{aligned}\boldsymbol{x}_s &= \left[y_{1,1,1,1},\ldots,y_{M,N,1,1},\ldots,y_{1,1,m',n'},\ldots,y_{M,N,m',n'},\ldots,y_{M,N,M,N}\right]^{\mathrm{T}} \\ &= \xi\boldsymbol{b}(\theta_0,\varphi_0)\otimes\boldsymbol{a}(R_0,\theta_0,\varphi_0)\end{aligned} \tag{9}$$

where $\boldsymbol{x}_s \in \mathbb{C}^{M^2N^2\times 1}$. $\boldsymbol{a}(R_0,\theta_0,\varphi_0)\in\mathbb{C}^{MN\times 1}$ and $\boldsymbol{b}(\theta_0,\varphi_0)\in\mathbb{C}^{MN\times 1}$ are the transmit and receive steering vectors, respectively, which can be expressed as

$$\boldsymbol{a}(R_0,\theta_0,\varphi_0) = \boldsymbol{a}_V(R_0,\varphi_0)\otimes\boldsymbol{a}_H(R_0,\theta_0,\varphi_0) \tag{10}$$

$$\boldsymbol{b}(\theta_0,\varphi_0) = \boldsymbol{b}_V(\varphi_0)\otimes\boldsymbol{b}_H(\theta_0,\varphi_0) \tag{11}$$

where $\boldsymbol{a}_V(R_0,\varphi_0)\in\mathbb{C}^{M\times 1}$, $\boldsymbol{a}_H(R_0,\theta_0,\varphi_0)\in\mathbb{C}^{N\times 1}$, $\boldsymbol{b}_V(\varphi_0)\in\mathbb{C}^{M\times 1}$ and $\boldsymbol{b}_H(\theta_0,\varphi_0)\in\mathbb{C}^{N\times 1}$ are the vertical transmit, horizontal transmit, vertical receive and horizontal receive steering vectors, respectively. Their specific representation forms are

$$\boldsymbol{a}_V(R_0,\varphi_0) = \left[1, e^{j2\pi\left(-2R_0\frac{\Delta f_{row}}{c}\right)},\ldots,e^{j2\pi\left(-2R_0\frac{(M-1)\Delta f_{row}}{c}\right)}\right]^{\mathrm{T}} \odot \left[1, e^{j2\pi\frac{d\sin(\varphi_0)}{\lambda_0}},\ldots,e^{j2\pi\frac{(M-1)d\sin(\varphi_0)}{\lambda_0}}\right]^{\mathrm{T}} \tag{12}$$

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$$\boldsymbol{a}_H\left(R_0,\theta_0,\varphi_0\right)=\left[1,e^{j2\pi\left(-2R_0\frac{\Delta f_{col}}{c}\right)},\ldots,e^{j2\pi\left(-2R_0\frac{(N-1)\Delta f_{col}}{c}\right)}\right]^{\mathrm{T}}\odot\left[1,e^{j2\pi\frac{d\sin(\theta_0)\cos(\varphi_0)}{\lambda_0}},\ldots,e^{j2\pi\frac{(N-1)d\sin(\theta_0)\cos(\varphi_0)}{\lambda_0}}\right]^{\mathrm{T}} \tag{13}$$

$$\boldsymbol{b}_V\left(\varphi_0\right)=\left[1,e^{j2\pi\frac{d\sin(\varphi_0)}{\lambda_0}},\ldots,e^{j2\pi\frac{(M-1)d\sin(\varphi_0)}{\lambda_0}}\right]^{\mathrm{T}} \tag{14}$$

$$\boldsymbol{b}_H\left(\theta_0,\varphi_0\right)=\left[1,e^{j2\pi\frac{d\sin(\theta_0)\cos(\varphi_0)}{\lambda_0}},\ldots,e^{j2\pi\frac{(N-1)d\sin(\theta_0)\cos(\varphi_0)}{\lambda_0}}\right]^{\mathrm{T}} \tag{15}$$

B. Signal model of mainlobe deceptive interference

We assume that there is a false target generator (FTG) located at (R_j',θ_j,φ_j), where R_j', θ_j and φ_j represent the slant range, azimuth angle and elevation angle of FTG. In this paper, the self-defense FTG loaded on the target is considered, i.e., $R_j'\approx R_0$, $\theta_j\approx\theta_0$ and $\varphi_j\approx\varphi_0$. According to Eq. (4), the signal seized by FTG is represented as

$$x_j\left(t\right)=\rho\sum_{m=1}^{M}\sum_{n=1}^{N}\alpha_{mn}\left(t-\frac{\tau_j}{2}-\tau_{j,m,n}\right)e^{j2\pi f_{mn}\left(t-\frac{\tau_j}{2}-\tau_{j,m,n}\right)} \tag{16}$$

where $\tau_j=2R_j'/c$ and $\tau_{j,m,n}=[d(m\text{-}1)\sin(\varphi_j)+d(n\text{-}1)\sin(\theta_j)\cos(\varphi_j)]/c$. Subsequently, the false target is generated through a process involving scaling, delaying, modulating, and retransmitting the captured signal. The jamming echo received by (m',n') receiving element is expressed as

$$y_{j,m',n'}\left(t\right)=\rho_j\sum_{m=1}^{M}\sum_{n=1}^{N}\alpha_{mn}\left(t-\tau_j-\tau_{j,m,n}-\tau_{j,m',n'}-\tau_{jam}\right)\times e^{j2\pi f_{mn}\left(t-\tau_j-\tau_{j,m,n}-\tau_{j,m',n'}-\tau_{jam}\right)} \tag{17}$$

where $\tau_{j,m',n}=[d(m'\text{-}1)\sin(\varphi_j)+d(n'\text{-}1)\sin(\theta_j)\cos(\varphi_j)]/c$. ρ_j denotes the complex-valued coefficient of false target. $\tau_{jam}=2\Delta r/c$ is the time delay caused by the FTG modulation. We use $R_j=R_j'+\Delta r$ to represent the equivalent range parameter of the false target. Essentially, adjusting the modulation delay enables changing the range of the false target.

After multi-channel matched filtering, the jamming echo corresponding to (m,n) transmitting element and (m',n') receiving element is obtained as

$$y_{j,m,n,m',n'}=\xi_j e^{j2\pi\left(-2R_j\frac{(m-1)\Delta f_{row}}{c}+\frac{(m-1)d\sin(\varphi_j)}{\lambda_0}\right)}e^{j2\pi\left(-2R_j\frac{(n-1)\Delta f_{col}}{c}+\frac{(n-1)d\sin(\theta_j)\cos(\varphi_j)}{\lambda_0}\right)}\times e^{j2\pi\frac{(m'-1)d\sin(\varphi_j)}{\lambda_0}}e^{j2\pi\frac{(n'-1)d\sin(\theta_j)\cos(\varphi_j)}{\lambda_0}} \tag{18}$$

where $\xi_j=\rho_j\exp\{-j4\pi f_0R_j/c\}$. Then, the snapshots of jamming for PFDA-MIMO radar can be written as

$$\begin{aligned}\boldsymbol{x}_j&=\left[y_{j,1,1,1,1},\ldots,y_{j,M,N,1,1},\ldots,y_{j,1,1,m',n'},\ldots,y_{j,M,N,m',n'},\ldots,y_{j,M,N,M,N}\right]^{\mathrm{T}}\\&=\xi_j\boldsymbol{b}\left(\theta_j,\varphi_j\right)\otimes\boldsymbol{a}\left(R_j,\theta_j,\varphi_j\right)\end{aligned} \tag{19}$$

where $\boldsymbol{x}_j\in\mathbb{C}^{M^2N^2\times1}$. Assuming that there are J false targets, the total data snapshots received by PFDA-MIMO radar is expressed as

$$\begin{aligned}\boldsymbol{x}&=\boldsymbol{x}_s+\boldsymbol{x}_j+\boldsymbol{n}_c\\&=\xi\boldsymbol{b}\left(\theta_0,\varphi_0\right)\otimes\boldsymbol{a}\left(R_0,\theta_0,\varphi_0\right)\\&\quad+\sum_{j=1}^{J}\xi_j\boldsymbol{b}\left(\theta_j,\varphi_j\right)\otimes\boldsymbol{a}\left(R_j,\theta_j,\varphi_j\right)+\boldsymbol{n}_c\end{aligned} \tag{20}$$

where $\boldsymbol{x}_s$, $\boldsymbol{x}_j$ and $\boldsymbol{n}_c$ are the snapshots of the true target, the false target and the Gaussian white noise, respectively.

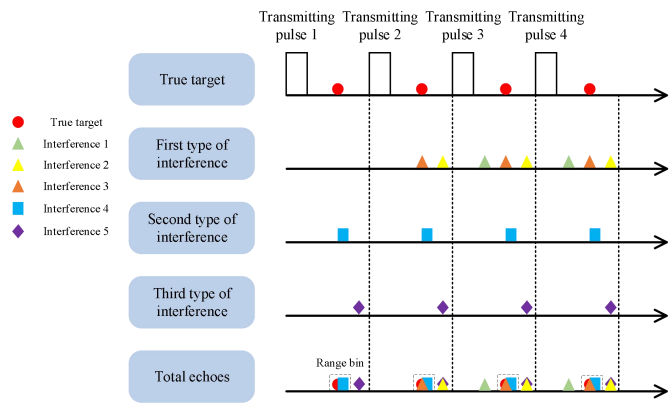

Fig. 4. Schematic representation of the received pulses from true and false targets.

Fig. 4 shows the schematic representation of the received pulses, containing echoes from both true and false targets. The red circles, green triangles, yellow triangles, orange triangles, blue rectangles and purple rhombus represent echoes from the true target and interference 1, 2, 3, 4, 5, respectively. The true target is located in the first range region while the mainlobe deceptive interference consists of the following three types.

1) Inter-pulse interference (interference 1, 2 and 3) is generated with a comparatively longer time delay, resulting in at least one pulse later than the true target. In other words, the range region index corresponding to the inter-pulse interference exceeds that of the true target. Note that although inter-pulse interference occurs in a different range region from the true target, within the same receiving window, its range bin may lead, lag, or even overlap with the true target in the same range bin. Specifically, the inter-pulse interference 1, preceding the true target, is located in the third range region, which is delayed by two pulses. The inter-pulse interference 2 is located in the second range region, positioned after the true target and delayed by one pulse. The inter-pulse interference 3 is located in the second range region and shares the same range bin with the true target.

2) Intra-pulse interference (interference 4), generated by fast modulation of FTG, shares the same number of delayed pulses and occupies the same range bin as the true target. Additionally, the time delay of the intra-pulse interference is slightly greater than that of the true target.

3) The interference shares the same number of delayed pulses but occupies the different range bin as the true target, as exemplified through interference 5. Although this type of interference is located in the same range region as the target, its range bin always lags behind that of the true target.

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In the following section, we will initially suppress inter-pulse interference in the horizontal spatial frequency domain. After suppressing inter-pulse interference, only the true target, the second and third types of interference remain in the echoes. Since the range bin of the third type interference is always behind that of the true target, within the same receiving window, the range bin of the true target definitely corresponds to the foremost range bin with high output power. Therefore, the third type of interference can be easily distinguished. In contrast, the second type of interference overlaps with the true target in the receiving window, making it impossible to directly distinguish. This paper focuses on suppressing the first and second types of interference, namely, inter-pulse and intra-pulse interferences joint suppression.

III. Mainlobe Deceptive Interference Suppression with PFDA-MIMO Radar

In this section, a two-stage beamforming method with PFDA-MIMO is introduced to suppress inter-pulse and intra-pulse interferences in joint horizontal-vertical spatial frequency domain.

A. Mainlobe deceptive inter-pulse interference suppression

In the PFDA-MIMO, the range information is incorporated into both horizontal and vertical transmit steering vectors. We define the horizontal and vertical transmit spatial frequency as

$$f_{TH}=-\frac{2R\Delta f_{col}}{c}+\frac{d\sin(\theta)\cos(\varphi)}{\lambda_0} \tag{21}$$

$$f_{TV}=-\frac{2R\Delta f_{row}}{c}+\frac{d\sin(\varphi)}{\lambda_0} \tag{22}$$

where $R=r+pR_u$. r, p (p=0,1,2,...) and R_u (R_u=c/(2f_{prf})) are the principal range, the delayed pulses number and the maximum unambiguous range, respectively. f_{prf} represents the pulse repetition frequency (PRF) of radar.

Although the range bin of inter-pulse interference may be positioned either in front of or behind the true target, or even same as that of the true target, the delayed pulses number of inter-pulse interference will invariably greater than that of the true target. To identify inter-pulse interference, horizontal transmit spatial frequency compensation (HTSFC) is performed in a range-bin-by-range-bin mode. The compensation vector is designed as

$$\boldsymbol{g}_{com1}=\mathbf{1}_{MN}\otimes\left(\mathbf{1}_M\otimes\boldsymbol{h}_{com1}(\hat{r}_l)\right) \tag{23}$$

where $\boldsymbol{g}_{com1}\in\mathbb{C}^{M^2N^2\times1}$, $\hat{r}_l$ (l=1,2,...,L) is the median range of the lth range bin with L is the number of total range bins. $\boldsymbol{h}_{com1}(\hat{r}_l)\in\mathbb{C}^{N\times1}$ is specifically given by

$$\boldsymbol{h}_{com1}(\hat{r}_l)=\left[1,e^{j4\pi\frac{\hat{r}_l\Delta f_{col}}{c}},\ldots,e^{j4\pi(N-1)\frac{\hat{r}_l\Delta f_{col}}{c}}\right]^{\mathrm{T}} \tag{24}$$

Then, compensation operation on the received data is performed as

$$\tilde{\boldsymbol{x}}=\boldsymbol{x}\odot\boldsymbol{g}_{com1} \tag{25}$$

Hence, the horizontal transmit spatial frequency of true target, intra-pulse interference and inter-pulse interference after HTSFC can be represented as

$$\tilde{f}_{TH_s}=-\frac{2r_{\Delta1}\Delta f_{col}}{c}+\frac{d\sin(\theta_0)\cos(\varphi_0)}{\lambda_0} \tag{26}$$

$$\tilde{f}_{TH_j_{intra\text{-}pulse}}=-\frac{2r_{\Delta2}\Delta f_{col}}{c}+\frac{d\sin(\theta_j)\cos(\varphi_j)}{\lambda_0} \tag{27}$$

$$\tilde{f}_{TH_j_{inter\text{-}pulse}}=-\left(\frac{2r_{\Delta3}\Delta f_{col}}{c}+\frac{2pR_u\Delta f_{col}}{c}\right)+\frac{d\sin(\theta_j)\cos(\varphi_j)}{\lambda_0},\ p\geq1 \tag{28}$$

where $r_{\Delta1}$, $r_{\Delta2}$ and $r_{\Delta3}$ are the range quantization errors of target, intra-pulse interference and inter-pulse interference, respectively. Since $\Delta f_{col}\ll B$, the horizontal transmit spatial frequency bias due to range quantization errors can be ignored. Thus, Eq. (26), (27) and (28) can be rewritten as

$$\tilde{f}_{TH_s}\approx\frac{d\sin(\theta_0)\cos(\varphi_0)}{\lambda_0} \tag{29}$$

$$\tilde{f}_{TH_j_{intra\text{-}pulse}}\approx\frac{d\sin(\theta_j)\cos(\varphi_j)}{\lambda_0} \tag{30}$$

$$\tilde{f}_{TH_j_{inter\text{-}pulse}}\approx-\frac{2pR_u\Delta f_{col}}{c}+\frac{d\sin(\theta_j)\cos(\varphi_j)}{\lambda_0},\ p\geq1 \tag{31}$$

After HTSFC, the true target and intra-pulse interferences share the identical horizontal transmit spatial frequency, while the inter-pulse interferences are distinct. By leveraging this characteristic, it becomes feasible to separate inter-pulse interferences from true target and intra-pulse interferences. Fig. 5 shows the spectral distribution of true and false targets after HTSFC. It can be seen that the true target and intra-pulse interference are clearly distributed diagonally in the horizontal transmit-receive spatial frequency domain. Additionally, since the true target and inter-pulse interferences occupy different range regions, there is a horizontal transmit spatial frequency difference of $-2pR_u\Delta f_{col}/c$ between them.

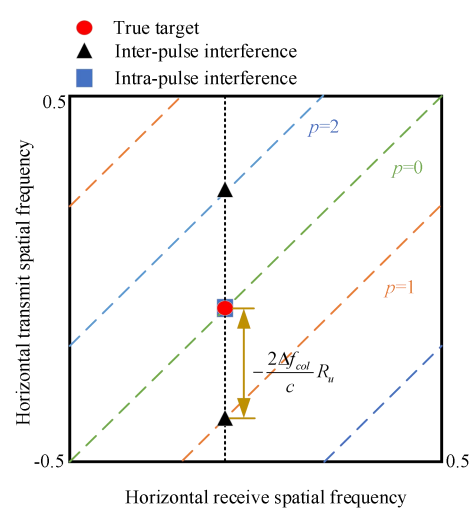

Fig. 5. Spectral distribution of true and false targets after HTSFC.

To ensure effective inter-pulse interference separation, Δf_{col} is necessary to be satisfied [39]

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$$\begin{cases} \dfrac{\Delta f_{col}}{f_{\text{prf}}} = u + q \\ qN_a \le 1 \end{cases} \tag{32}$$

where $u \in \mathbb{N}$ and $q \in (0,1)$ denote the non-negative integral portion and the decimal portion, respectively. N_a is the number of total range regions.

Due to the specific distribution of the target and interferences in the horizontal transmit-receive spatial frequency domain, non-adaptive beamforming is employed. The non-adaptive transmit beampattern of a uniform array with N elements possesses N-1 nulls, with the spatial frequency interval between adjacent nulls being $1/N$. Hence, to ensure that inter-pulse interferences are distributed at the nulls of the beampattern, it is also imperative to satisfy

$$qp = \frac{i}{N}, \quad i = 1, 2, \ldots, N-1 \tag{33}$$

When Δf_{col} satisfies both Eq. (32) and (33), horizontal transmit-receive non-adaptive beamforming effectively suppresses inter-pulse interference while maximizing the gain at the target. The output can be represented as

$$\boldsymbol{y} = \left\{ \left[\mathbf{I}_M \otimes \boldsymbol{b}_H(\theta_0, \varphi_0) \right] \otimes \left[\mathbf{I}_M \otimes \tilde{\boldsymbol{a}}_H(\tilde{f}_{TH_s}) \right] \right\}^{\mathrm{H}} \tilde{\boldsymbol{x}} \tag{34}$$

where $\boldsymbol{y} \in \mathbb{C}^{M^2 \times 1}$. $\tilde{\boldsymbol{a}}_H(\tilde{f}_{TH_s})$ denotes the horizontal transmit steering vector of the target after HTSFC and is represented as

$$\tilde{\boldsymbol{a}}_H(\tilde{f}_{TH_s}) = \left[1, e^{j2\pi \tilde{f}_{TH_s}}, \ldots, e^{j2\pi(N-1)\tilde{f}_{TH_s}} \right]^{\mathrm{T}} \tag{35}$$

B. Mainlobe deceptive intra-pulse interference suppression

After suppressing inter-pulse interferences, the target must be located in the range bin that initially exhibits high output power (later referred to as the target range bin). The interferences that share the identical range region but occupy different range bins from the true target (the third type of interference) can be easily distinguished and will not affect target detection after the first-stage beamforming. Therefore, this type of interference is not considered further in this part. In contrast, the suppression of intra-pulse interference, which shares the same range region and range bin as the true target, is the focus of this part. Since the equivalent range of intra-pulse interference is consistently greater than that of the true target, i.e., R_j-R_0=ΔR>0., the true target can be distinguished in vertical transmit spatial frequency domain by exploiting the range DOFs.

To ensure that the true target and intra-pulse interferences are distributed according to a fixed pattern, the vertical transmit spatial frequency compensation (VTSFC) is conducted. The compensation vector is constructed as

$$\boldsymbol{g}_{com2} = \mathbf{1}_M \otimes \boldsymbol{h}_{com2}(\bar{r}_s) \tag{36}$$

where $\boldsymbol{g}_{com2} \in \mathbb{C}^{M^2 \times 1}$. $\bar{r}_s$ is the initial range of the target range bin. $\boldsymbol{h}_{com2}(\bar{r}_s) \in \mathbb{C}^{M \times 1}$ is further expressed as

$$\boldsymbol{h}_{com2}(\bar{r}_s) = \left[1, e^{j4\pi \frac{\bar{r}_s \Delta f_{row}}{c}}, \ldots, e^{j4\pi(M-1)\frac{\bar{r}_s \Delta f_{row}}{c}} \right]^{\mathrm{T}} \tag{37}$$

The echo after VTSFC is represented as

$$\tilde{\boldsymbol{y}} = \boldsymbol{y} \odot \boldsymbol{g}_{com2} \tag{38}$$

Consequently, the vertical transmit spatial frequency of true target and intra-pulse interference can be rewritten as

$$\tilde{f}_{TV_s} = -\frac{2(R_0 - \bar{r}_s)\Delta f_{row}}{c} + \frac{d \sin(\varphi_0)}{\lambda_0} \tag{39}$$

$$\tilde{f}_{TV_j_{intra\text{-}pulse}} = -\frac{2(R_j - \bar{r}_s)\Delta f_{row}}{c} + \frac{d \sin(\varphi_j)}{\lambda_0} \tag{40}$$

The spectral distribution of the true target and intra-pulse interferences after VTSFC is illustrated in Fig. 6. The red circle, green star and blue squares represent the true target, the beginning of target range bin and intra-pulse interferences, respectively. Note that the vertical transmit and receive spatial frequencies of the points on the green dashed line are both $d\sin(\varphi)/\lambda_0$. It is evident that the beginning of target range bin is arranged diagonally in the vertical transmit-receive spatial frequency domain. Within this range bin, as range increases, the vertical transmit spatial frequency will change along the direction of the purple arrow, i.e., from $d\sin(\varphi)/\lambda_0$ to -0.5, and then from 0.5 to $d\sin(\varphi)/\lambda_0$. Due to R_0<R_j, along the direction of the purple arrows, the true target always lags behind the intra-pulse interference in the vertical transmit spatial frequency domain with a frequency difference of $-2\Delta R\Delta f_{row}/c$. It is important to highlight that while the positions of the target and intra-pulse interferences in the vertical transmit spatial frequency domain are not fixed, their relative positional relationship can remain constant.

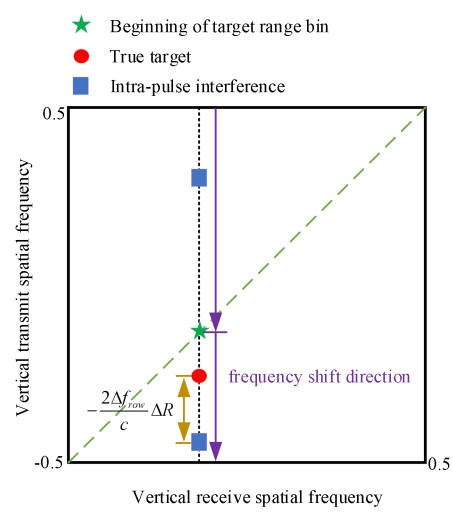

Fig. 6. Spectral distribution of true target and intra-pulse interferences after VTSFC.

To ensure the target consistently stays behind of the intra-pulse interferences, the variation of vertical transmit spatial frequency needs to be no greater than 1. Thus, Δf_{row} should be satisfied

$$\Delta f_{row} \le \frac{c}{2r_{\text{bin}}} \tag{41}$$

where r_{bin} is the width of a range bin. It is particularly emphasized that the range between true target and intra-pulse interference is close. Thus, Δf_{row} should be as large as possible to ensure an obvious separation.

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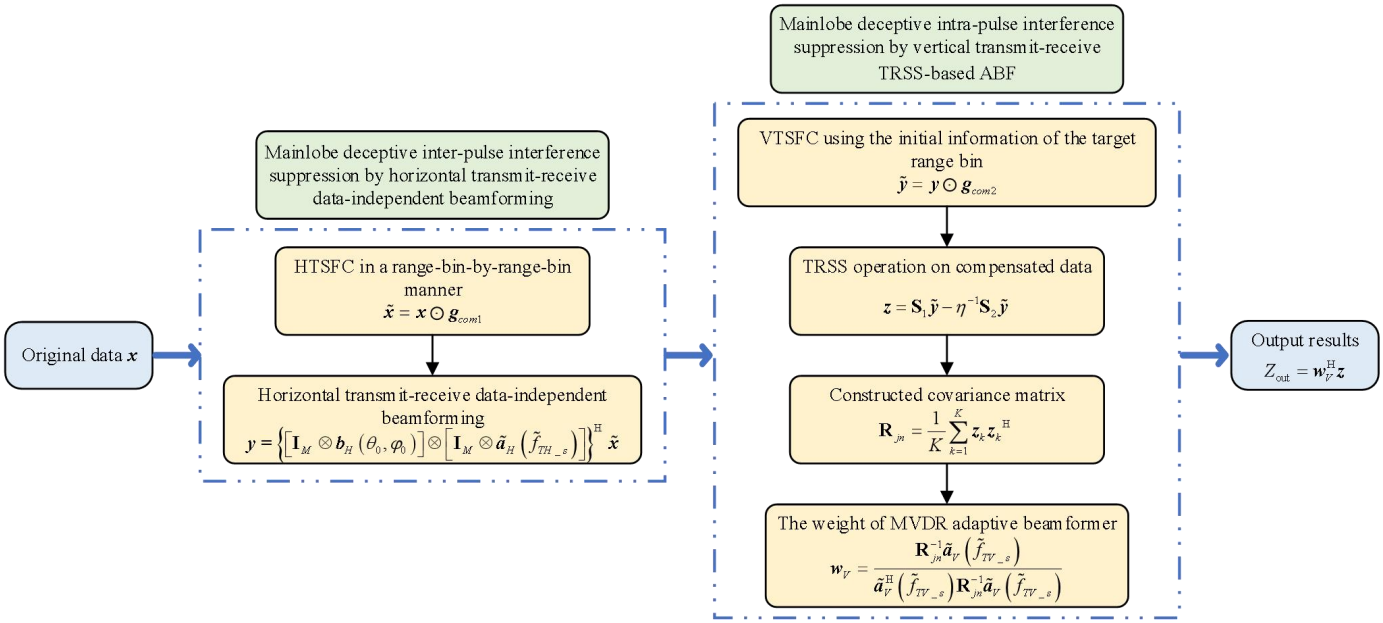

Fig. 7. Flow chart of two-stage mainlobe deceptive interference suppression method with PFDA-MIMO radar.

Subsequently, vertical transmit-receive ABF is performed to mitigate intra-pulse interferences. As the presence of target signal in the training samples, conventional minimizing variance distortionless response (MVDR) ABF method will lead to target cancellation. To this end, an ABF method based on TRSS is introduced. TRSS operation is conducted on compensated data, which can be written as

$$\boldsymbol{z} = \mathbf{S}_1\tilde{\boldsymbol{y}} - \eta^{-1}\mathbf{S}_2\tilde{\boldsymbol{y}} \tag{42}$$

where η=exp $\{j2\pi\tilde{f}_{TV_s}\}$. The selection matrix $\mathbf{S}_1$ and $\mathbf{S}_2$ are denoted as

$$\mathbf{S}_1 = \mathbf{I}_M \otimes \left[\mathbf{I}_{M-1}, \mathbf{0}_{M-1}\right] \tag{43}$$

$$\mathbf{S}_2 = \mathbf{I}_M \otimes \left[\mathbf{0}_{M-1}, \mathbf{I}_{M-1}\right] \tag{44}$$

After TRSS operation, the covariance matrix only contains intra-pulse interferences and noise (INM), which can be estimated as

$$\mathbf{R}_{jn} = \frac{1}{K}\sum_{k=1}^{K} \boldsymbol{z}_k \boldsymbol{z}_k^{\mathrm{H}} \tag{45}$$

where $\boldsymbol{z}_k$ is the snapshot of the kth pulse. The weight of MVDR adaptive beamformer can be formulated as

$$\boldsymbol{w}_V = \frac{\mathbf{R}_{jn}^{-1}\tilde{\boldsymbol{a}}_V\left(\tilde{f}_{TV_s}\right)}{\tilde{\boldsymbol{a}}_V^{\mathrm{H}}\left(\tilde{f}_{TV_s}\right)\mathbf{R}_{jn}^{-1}\tilde{\boldsymbol{a}}_V\left(\tilde{f}_{TV_s}\right)} \tag{46}$$

where $\tilde{\boldsymbol{a}}_V(\tilde{f}_{TV_s})$ denote the vertical transmit steering vector of the target after VTSFC, which specific form is

$$\tilde{\boldsymbol{a}}_V\left(\tilde{f}_{TV_s}\right) = \left[1, e^{j2\pi\tilde{f}_{TV_s}}, \ldots, e^{j2\pi(M-1)\tilde{f}_{TV_s}}\right]^{\mathrm{T}} \tag{47}$$

Fig. 7 summarizes the proposed two-stage mainlobe deception interference suppression method with the PFDA-MIMO radar. Firstly, HTSFC is applied in horizontal transmit spatial frequency domain on a range-bin-by-range-bin basis. Through judicious design of Δf_{col}, inter-pulse interferences can be distinguished from the true target and intra-pulse interferences. Subsequently, horizontal transmit-receive data-independent beamforming is employed to mitigate inter-pulse interference. In the second stage, VTSFC is performed using the initial range of the target range bin. By appropriately adjusting Δf_{row}, the true target clearly lags behind the intra-pulse interferences in the vertical transmit spatial frequency domain. Finally, vertical transmit-receive TRSS-based ABF is introduced to suppress intra-pulse interference.

IV. Experimental Results and Discussions

In this section, simulations are carried out to assess the validity of the proposed mainlobe deceptive interference suppression method with PFDA-MIMO radar. Moreover, the performance of the proposed method is also in comparison with that of MIMO radar (Δf_{row}=0, Δf_{col}=0) and FDA-MIMO radar (Δf_{row}=0, Δf_{col}=10kHz). The concrete values of radar system parameters are provided in Table I. Assuming that there are two intra-pulse interferences (P_1, P_2), three inter-pulse interferences (P_3, P_4, P_5) and one true target P_0 in the mainlobe. Multiple specific parameters are displayed in Table II.

TABLE I
Parameters of Radar System

Symbol	Parameters	Values
M	Row numbers of the planar array	9
N	Column numbers of the planar array	9
f_0	Reference carrier frequency	1GHz
f_{prf}	Pulse repetition frequency	7500Hz
Δf_{row}	Row frequency increment	1.4MHz
Δf_{col}	Column frequency increment	10kHz
d	Planar array element spacing	0.15m
N_a	Number of total range regions	3
r_{bin}	Width of a range bin	100m
L	Number of range bins	200

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TABLE II
MULTIPLE SPECIFIC PARAMETERS OF TRUE AND FALSE TARGETS

	Target P_0 (true)	Target P_1 (false)	Target P_2 (false)	Target P_3 (false)	Target P_4 (false)	Target P_5 (false)
Azimuth angle(°)	5	5	5	5	5	5
Elevation angle(°)	5	5	5	5	5	5
Delayed pulses	0	0	0	1	1	2
SNR/JNR(dB)	5	30	30	30	30	30
Index of range bin	41	41	41	41	141	31
Equivalent range (km)	10.02	10.05	10.08	30	40	49
Horizontal transmit spatial frequency of PFDA-MIMO	0.3754	0.3734	0.3714	0.0434	0.3767	-0.2233
Horizontal receive spatial frequency of PFDA-MIMO	0.0434	0.0434	0.0434	0.0434	0.0434	0.0434
Vertical transmit spatial frequency of PFDA-MIMO	-0.4764	0.2436	-0.0364	0.0436	-0.2898	-0.2898
Vertical receive spatial frequency of PFDA-MIMO	0.0436	0.0436	0.0436	0.0436	0.0436	0.0436

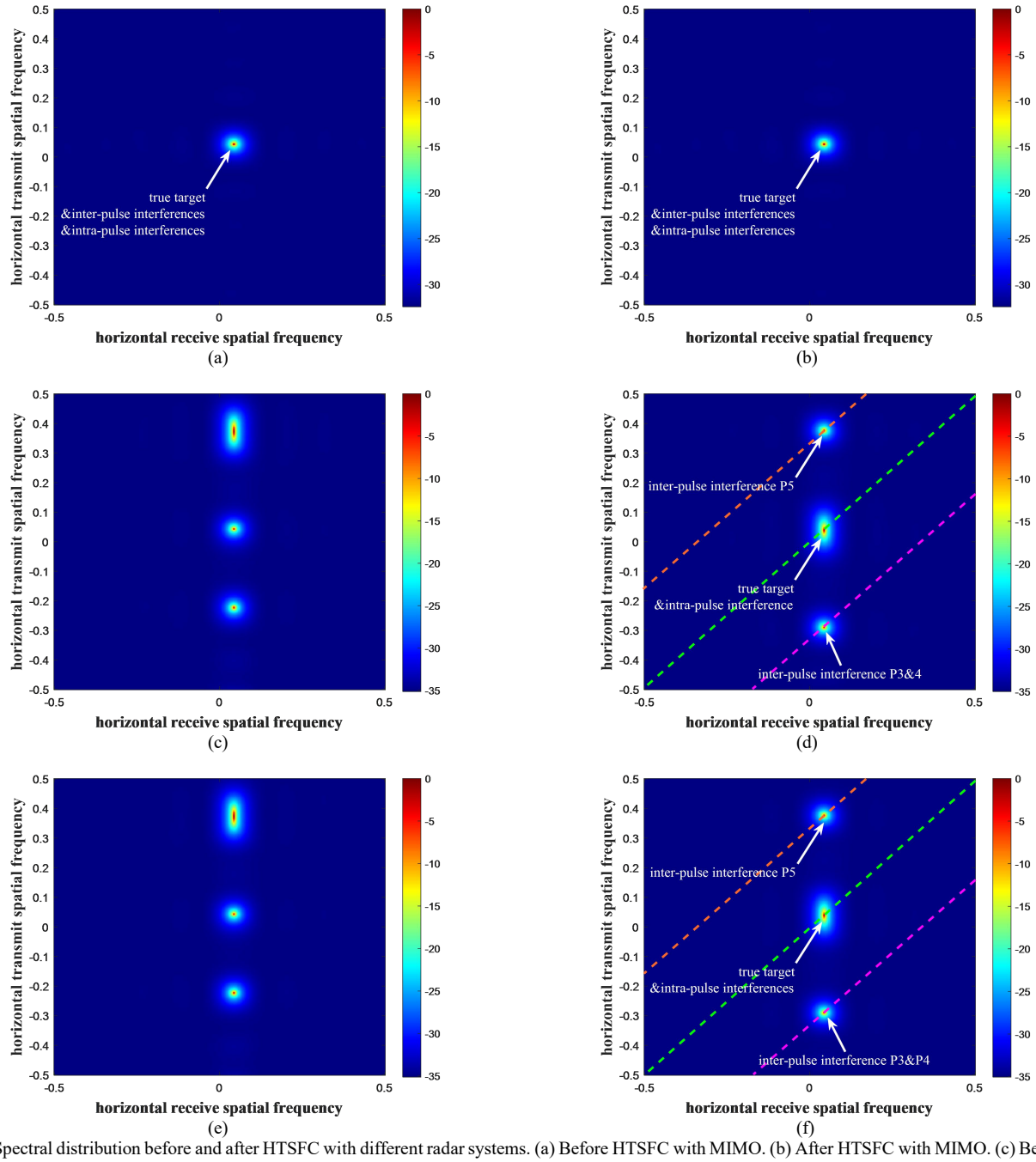

Fig. 8. Spectral distribution before and after HTSFC with different radar systems. (a) Before HTSFC with MIMO. (b) After HTSFC with MIMO. (c) Before HTSFC with FDA-MIMO. (d) After HTSFC with FDA-MIMO. (e) Before HTSFC with PFDA-MIMO. (f) After HTSFC with PFDA-MIMO.

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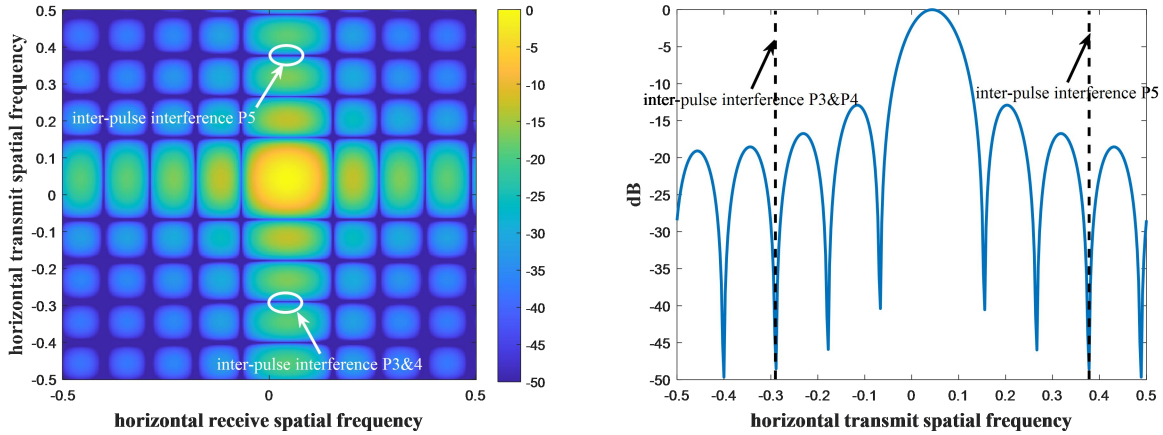

Fig. 9. Non-adaptive beamformer response of PFDA-MIMO radar. (a) Horizontal transmit-receive spatial frequency domain. (b) Slice of horizontal transmit spatial frequency domain.

Fig. 8 presents the spectral distribution results for MIMO, FDA-MIMO and PFDA-MIMO, all before and after HTSFC. Since the steering vector of MIMO radar lacks range DOFs, it cannot distinguish between the true target and interferences, as illustrated in Fig. 8(a) and (b). The horizontal steering vectors of FDA-MIMO and PFDA-MIMO are identical, resulting in the same spectral distributions in the horizontal spatial frequency domain, as depicted in Fig. 8(c)-(f). From Fig. 8(c) and (e), it can be observed that the spectral distributions of target and interferences before HTSFC are irregular. After HTSFC, the true target and intra-pulse interferences have nearly identical horizontal transmit spatial frequency and exhibit diagonal distribution in the horizontal transmit-receive spatial frequency domain. However, the distributions of inter-pulse interferences differ from that of the true target, as shown in Fig. 8(d) and (f). Hence, both FDA-MIMO and PFDA-MIMO can suppress inter-pulse interference through beamforming.

After HTSFC, the true target and interferences have specific distributions in the horizontal spatial frequency domain. Therefore, non-adaptive beamforming, which does not require training samples and reduces computational complexity, is employed to suppress inter-pulse interferences. Fig. 9 shows the non-adaptive beamformer response of PFDA-MIMO in horizontal spatial frequency domain. Fig. 9(b) is a slice of Fig. 9(a) with the horizontal receive spatial frequency of 0.0434. It can be evident that when Δf_{col} is designed according to Eq. (33), inter-pulse interferences (P_3, P_4 and P_5) are all located at the nulls of the beampattern and can be well suppressed.

After the horizontal transmit-receive beamforming, the spectral distribution in vertical spatial frequency domain, before and after VTSFC, for MIMO, FDA-MIMO and PFDA-MIMO is shown in Fig. 10. Since the vertical steering vectors of MIMO radar lacks range DOFs, it cannot effectively separate both inter-pulse interferences and intra-pulse interferences. The spectra of target, inter-pulse interferences and intra-pulse interferences overlap in the vertical spatial frequency domain, as shown in Fig. 10(a) and (b). Although the horizontal transmit steering vector of FDA-MIMO depends on range, its vertical transmit steering vector is independent of range. Therefore, while FDA-MIMO can suppress inter-pulse interference through horizontal transmit-receive beamforming, it cannot differentiate intra-pulse interferences from target, as shown in Fig. 10(c) and (d). In contrast to FDA-MIMO, PFDA-MIMO incorporates additional range DOFs in both vertical and horizontal transmit steering vectors. By leveraging this feature, intra-pulse interferences can be distinguished from the true target in the vertical transmit spatial frequency domain. Before VTSFC, the spectral distribution of the true target and intra-pulse interferences are irregular, as shown in Fig. 10(e). Fig. 10(f) shows the spectral distribution of PFDA-MIMO with Δf_{row} satisfying Eq. (41) after VTSFC. The green circle indicates the beginning of the target range bin. Along the direction of the purple arrows, the true target will lag behind the intra-pulse interferences and exhibit a significant separation effect, thus identifying the true target.

The vertical transmit-receive adaptive beamformer response of PFDA-MIMO is demonstrated in Fig. 11. As a comparison, Fig. 11(a), (b) and (c) correspond to the responses of the conventional, TRSS-based and optimal MVDR beamformers, respectively. Fig. 11(d) represents slices of the three beamformer responses in the vertical transmit spatial frequency domain. Due to the presence of target component in the covariance matrix of echo, the spatial response of the traditional MVDR adaptive beamformer is distorted, as shown in Fig. 11(a). Consequently, the method based on traditional MVDR adaptive beamforming fails to effectively suppress interference, as illustrated by the blue line in Fig. 11(d). In contrast, the TRSS-based MVDR beamformer proposed in this paper effectively addresses the issue of target cancellation, aligning the mainlobe with the true target and generating nulls at the positions of intra-pulse interferences, as demonstrated in Fig. 11(b). The optimal MVDR beamformer requires the accurate INM, which provides the best interference suppression effect, as shown in Fig. 11(c). However, in practice, the accurate INM is not available. Compared to the optimal MVDR beamformer, the TRSS-based MVDR beamformer achieves approximate interference suppression result at the cost of sacrificing an effective aperture, as shown in Fig. 11(d). However, since the performance loss is minimal, this trade-off is acceptable.

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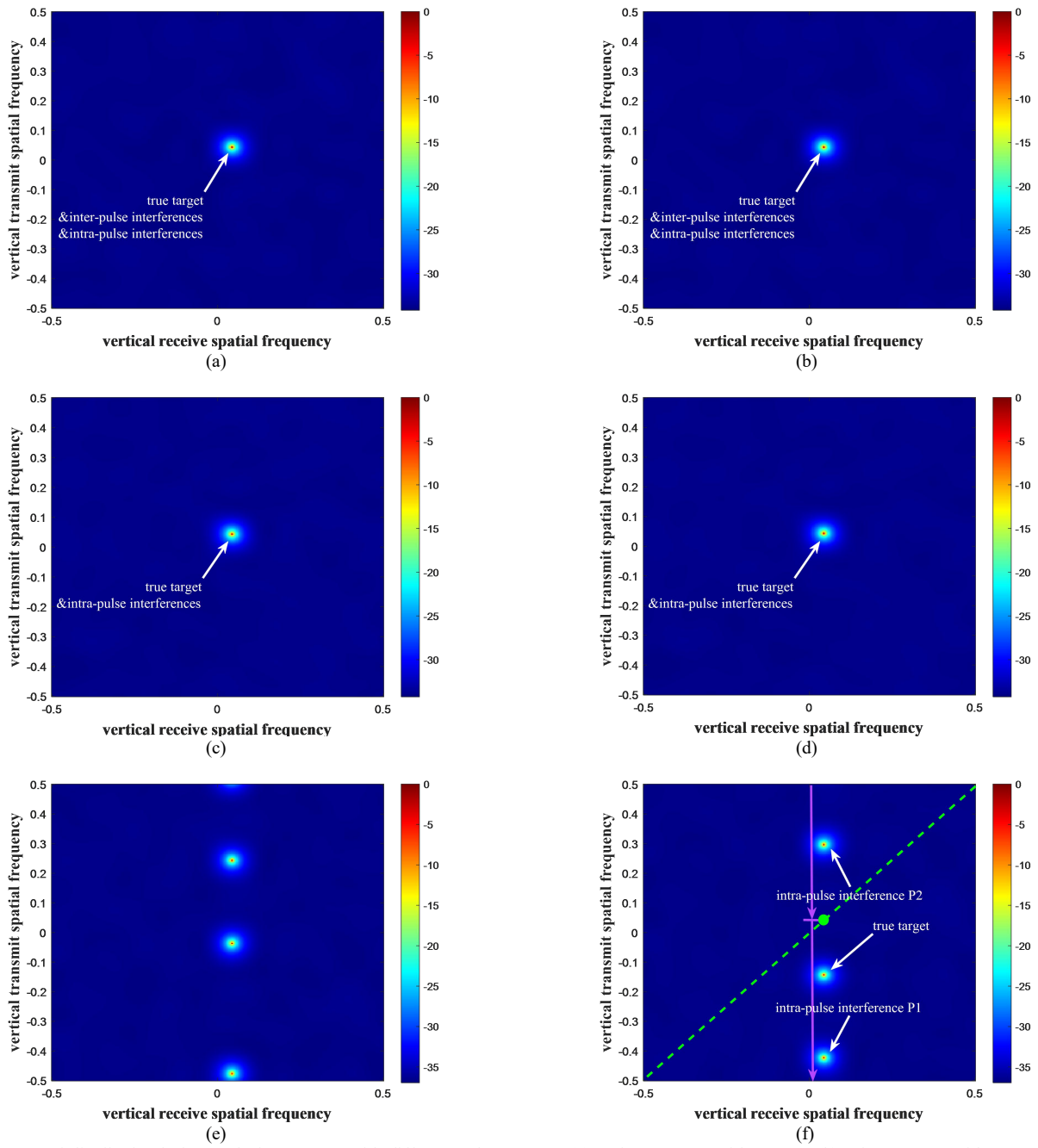

Fig. 10. Spectral distribution before and after VTSFC with different radar systems. (a) Before VTSFC with MIMO. (b) After VTSFC with MIMO. (c) Before VTSFC with FDA-MIMO. (d) After VTSFC with FDA-MIMO. (e) Before VTSFC with PFDA-MIMO. (f) After VTSFC with PFDA-MIMO.

Furthermore, the performance comparisons of different radar systems are presented in Fig. 12. Fig. 12(a) shows the output power of MIMO, FDA-MIMO and PFDA-MIMO within 200 range bins. The true target P_0 and intra-pulse interferences P_1 and P_2 are all located at the first range region corresponding to the 41th range bin. Inter-pulse interferences P_3 and P_4 are located at the second range region, corresponding to the 41th range bin and 141th range bin, respectively. Inter-pulse interference P_5 is located in the third range region, corresponding to the 31th range bin. Although their range region indices are distinct, owing to range ambiguity, they will appear simultaneously in the same receiving window. The ideal performance is displayed by a black dashed line for comparison. MIMO radar is unable to effectively suppress mainlobe interference, leading to high output power for both intra-pulse and inter-pulse interferences, as depicted by the blue line. FDA-MIMO can suppress inter-pulse interferences through first-stage beamforming, but it cannot distinguish intra-pulse interferences from true target, resulting in high output power for intra-pulse interferences, as shown by the red line. PFDA-MIMO can effectively suppress both intra-pulse and inter-pulse interferences, approaching ideal performance, as indicated by the green line. The variations of the output signal-to-interference-plus-noise ratio (SINR) with respect to

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input SNR for different radars are shown in Fig. 12(b). For comparison, the 10lg(M^2N^2) gain is shown as the upper bound, represented by a black dashed line. It is evident that MIMO radar struggles to effectively separate the mainlobe interferences, resulting in a significant degradation in interference suppression performance. While FDA-MIMO can suppress mainlobe inter-pulse interference, it fails to mitigate intra-pulse interferences, leading to a decrease in system performance. In contrast, PFDA-MIMO achieves the highest output SINR, with its performance falling no more than 3dB below the upper bound.

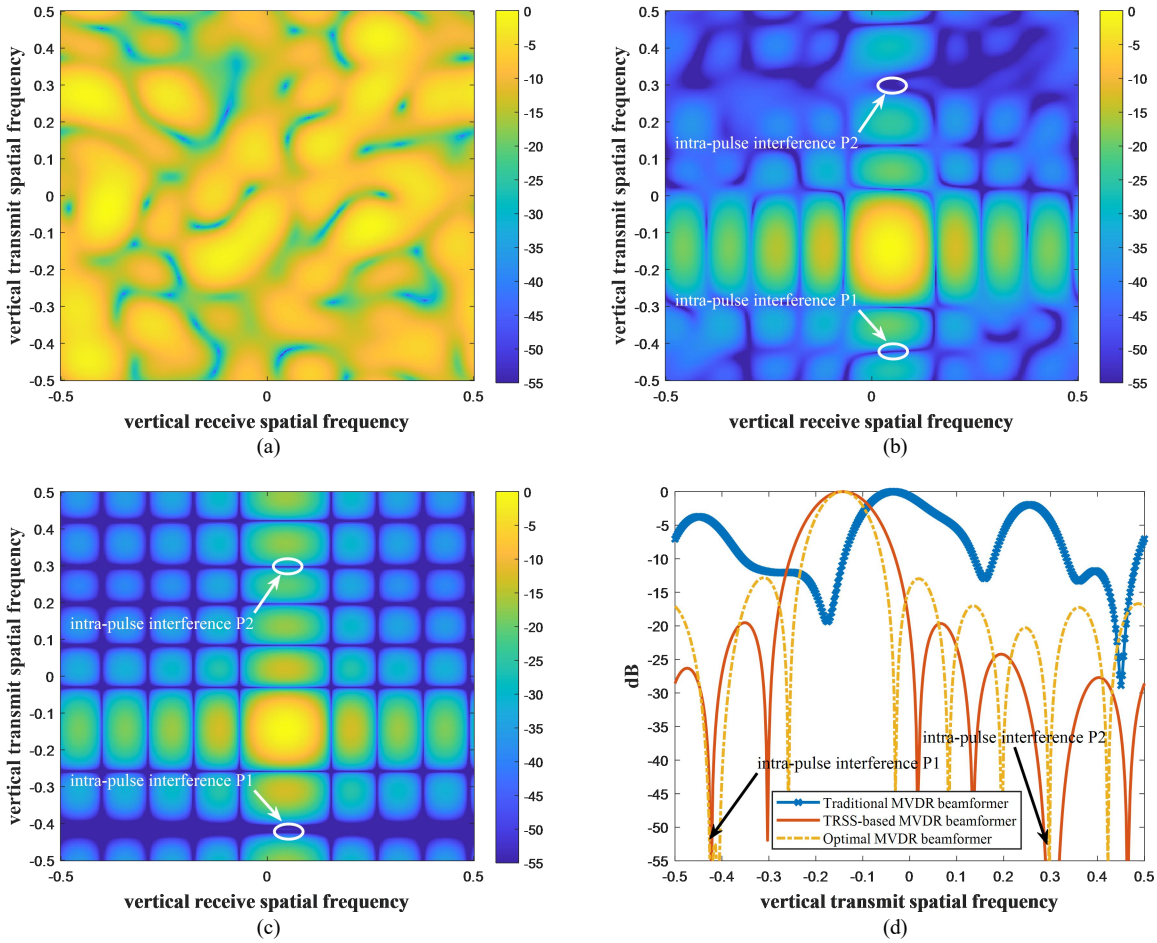

Fig. 11. Vertical transmit-receive adaptive beamformer response of PFDA-MIMO radar. (a) Traditional MVDR beamformer. (b) TRSS-based MVDR beamformer. (c) Optimal MVDR beamformer (d) Slices of vertical transmit spatial frequency domain.

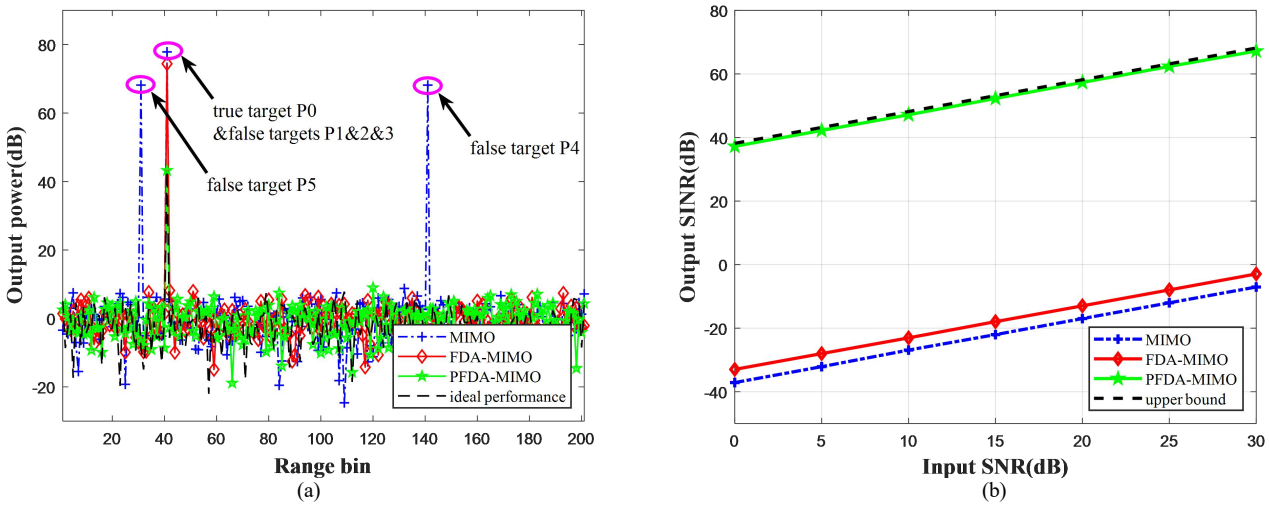

Fig. 12. Performance comparisons with different radars. (a) Output power. (b) Output SINR versus input SNR.

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V. Conclusion

This paper presents a PFDA-MIMO framework and introduces a two-stage beamforming method for mainlobe deceptive interference suppression. Firstly, the signal model of PFDA-MIMO is derived, and multiple matched filters are designed to recover the DOFs in transmitter. Different from FDA-MIMO, PFDA-MIMO incorporates controllable 2-D range DOFs in both the vertical and horizontal transmit steering vectors, enabling the differentiation between the true target, intra-pulse interferences, and inter-pulse interferences. Subsequently, a two-stage beamforming method is introduced to jointly suppress deceptive inter-pulse and intra-pulse interferences: 1) horizontal transmit-receive data-independent beamforming for suppressing inter-pulse interferences; 2) vertical transmit-receive ABF based on TRSS for suppressing intra-pulse interferences. Moreover, the frequency increments in both rows and columns of planar array are reasonably designed to ensure effective interferences separation. Simulation results demonstrate that PFDA-MIMO performs excellently in suppressing interference when deceptive inter-pulse and intra-pulse interferences coexist.

Future studies will extend PFDA to tasks such as target parameter estimation and clutter suppression. It is noted that cross-correlation of imperfect orthogonal waveforms and the spatial frequency mismatch can lead to a decrease in actual performance, which will be addressed in future research.

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